Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18EC53]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

DIGITAL SIGNAL PROCESSING

Time:3 Hours **Maximum Marks:70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

1.	**a.**	Define DFT and IDFT of a signal. Obtain the relationship between DFT and Z transform	**06**
	b.	Determine 8 point DFT of the signal x(n)={1,1,1,1,1,1,0,0}	**08**
2.	**a.**	Use overlap save method to compute y(n) of FIR filter with impulse response h(n)={3,2,1} and input x(n)=(2,1,-1,-2,-3,5,6,-1,2,0,2,1).Use only 8 point circular convolution in your approach.	**08**
	b.	Determine 4 point IDFT of X(K)={2.5,-0.25+j0.75,0,-0.25-j0.75}	**06**
3.	**a.**	Determine the expression for the order, cutoff frequency and poles of butterworth filter.	**08**
	b.	Transform $H(s)=\frac{(s+a)}{(s+a)^2+b^2}$ into a digital filter using impulse invariance method. Use T=1 Sec	**06**

OR

4.	**a.**	Design a unit bandwidth 3-dB digital Butterworth filter of order one by using bilinear transformation.	**06**
	b.	Design a digital low pass filter using the bilinear transformation method to satisfy the following characteristics: **i.** Monolithic stop band and pass band **ii.** -3db cut off frequency of 0.5π rad **iii.** Magnitude down at least 15 dB at 0.75π rad	**08**
5.		Design a Chebyshev 1 filter to meet the following specifications: **i.** Pass band ripple : ≤2dB **ii.** Pass band edge : 1 rad/sec **iii.** Stop band attenuation: ≥20 dB **iv.** Stop band edge:1.3 rad/sec	**14**

OR

6. Design a digital Chebyshev 1 filter that satisfies the following constraints. **14**

$0.8 \le |H(\omega)| \le 1, \quad 0 \le \omega \le 0.2\pi$

$|H(\omega)| \le 0.2, \quad 0.6\pi \le \omega \le \pi$

7. a. Obtain the cascade realization of the following H(Z) **04**

$H(z)=\frac{8z^3-4z^2+11z-2}{(z-0.25)(z^2-z+0.5)}$

b. The frequency response of a desired low pass filter is given by **10**

$$H_d(\omega) = \begin{cases} e^{-j2\omega} & ; -\pi/4 \le |\omega| \le \pi/4 \\ 0 & ; \pi/4 < |\omega| \le \pi \end{cases}$$

Use rectangular window. Obtain or develop the frequency response expression.

OR

8. a. Realize the following system function $H(z)=1+\frac{3}{4}z^{-1}+\frac{17}{8}z^{-2}+\frac{3}{4}z^{-3}+z^{-4}$ in **06**

i. Direct form

ii. Cascaded form

b. Explain the following windows **08**

i. Rectangular window

ii. Hamming Window

iii. Blackman Window

iv. Hanning window